

Customer Churn Analysis

An end-to-end churn analysis project on a 50,300-record customer dataset — cleaning and transforming raw data in Python, building business KPIs in SQL Server, and visualizing churn drivers in Power BI.

Overview

Customer churn directly impacts recurring revenue, and most churn datasets arrive with missing values, mixed data types, and no pre-built business logic. This project cleans a raw customer churn dataset in Python, loads it into SQL Server for KPI-level querying, and uses Power BI to turn the output into a churn dashboard that surfaces which customer segments, regions, and satisfaction levels are driving churn.

Problem Statement

The raw dataset had data quality gaps that blocked direct analysis: - Missing values in Age, Region, AvgOrderValue, SatisfactionScore, and NPSScore - ChurnReason populated only for churned customers, blank for the rest - Date fields (JoinDate, LastPurchaseDate) stored as text in DD-MM-YYYY format - 300 duplicate records - No segmentation, churn KPIs, or satisfaction/NPS breakdowns pre-built for reporting

The goal was to clean the data, calculate churn and satisfaction KPIs at the SQL layer, and answer: **which customers are churning, and what separates them from the customers who stay?**

Dataset

- **File:** churn_analysis.csv
- **Size:** 50,300 rows × 17 columns (before cleaning)
- **Columns:** CustomerID, JoinDate, LastPurchaseDate, Age, Gender, Region, CustomerSegment, ProductCategory, TotalPurchases, TotalSpend, AvgOrderValue, DaysSinceLastPurchase, NumSupportTickets, SatisfactionScore, NPSScore, Churn, ChurnReason

Tools and Technologies

- **Language:** Python
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, SQLAlchemy
- **Database:** Microsoft SQL Server (via SSMS)
- **Dashboard:** Power BI (CHURN.pbix)
- **Environment:** Jupyter Notebook

Methods

1. Data Cleaning (Python) - Imputed missing Age with the median - Imputed missing Region with the mode - Imputed missing AvgOrderValue with the median, grouped by CustomerSegment and TotalPurchases, to keep the imputed value consistent with similar customers rather than a single global median - Imputed missing NPSScore and SatisfactionScore with the median - Set ChurnReason to "Not_Churned" wherever Churn = 0, so every row has a populated reason field - Converted JoinDate and LastPurchaseDate from text to proper datetime format - Converted Age from float to integer - Removed 300 duplicate records

2. Feature Engineering - Age_Group: customers segmented into Young, Adult, Senior using equal-frequency binning (pd.qcut)

3. Exploratory Data Analysis (Python) - Outlier detection via boxplots across Age, TotalPurchases, TotalSpend, SatisfactionScore, NPSScore - Gender distribution (pie chart) and gender vs. total spend (bar chart) - Gender vs. churn (clustered bar chart) - Age vs. average order value (scatter plot) to check for a relationship - Total purchases by gender (line chart) - Correlation heatmap across all numeric features - Positive vs. negative review counts by customer segment

4. SQL Server Analysis The cleaned dataset was loaded into SQL Server (table churn_info) via SQLAlchemy, then queried to produce business-ready KPIs (SQLQuery.sql): - Total sales, and sales broken down by age group, region, product category, and gender - Average satisfaction score by age group - Average satisfaction score split by churned vs. not-churned customers, overall and by age group - NPS category counts (Bad / Average / Excellent / World-class) by age group - Product category sales as a percentage of total sales - Average days since last purchase by region and age group - Sales by customer segment, and by region × customer segment

5. Dashboard - CHURN.pbix: Power BI dashboard built on the cleaned dataset and SQL-layer KPIs for interactive exploration of churn by segment, region, and satisfaction.

Key Insights

- **Scale:** 50,000 unique customers after removing 300 duplicates (one record per customer — this dataset has no repeat-row customers to begin with).
- **Churn rate:** 30.65% of customers have churned; 69.35% are retained.
- **Gender split:** Female customers make up 50.1% of the base, Male 47.9%, Other 2.0% — a near-even split, not a skewed one.
- **Spend by gender:** Female customers contribute the highest total spend, followed closely by Male customers; the “Other” category contributes a small fraction of total spend, consistent with its smaller population share.
- **Age vs. spend behavior:** The scatter plot of Age vs. AvgOrderValue shows no visible trend — age does not appear to predict order value in this dataset.
- **Correlation with churn:** Satisfaction Score (-0.23) and NPS Score (-0.18) both have a weak negative correlation with churn — more satisfied customers churn somewhat less, but the relationship is weak, not a strong predictor on its own.
- **Other correlations:** TotalPurchases and AvgOrderValue have a moderate negative correlation (-0.43) — customers with more purchases tend to have a lower value per order. TotalSpend and AvgOrderValue have a weak positive correlation (0.23).
- **Segment revenue:** SMB is the highest-revenue customer segment, followed by Mid-Market, Startup, and Enterprise.
- **Product category revenue:** Subscription is the top-selling category, followed by Software, Consulting, Services, and Hardware.
- **Regional revenue:** Central region leads in total sales, though the gap between Central, West, South, East, and North is relatively narrow — no single region dominates.

Note on scope: The correlations above are weak-to-moderate at best. Satisfaction and NPS being only weakly correlated with churn means this dataset alone isn't enough to build a reliable churn-prediction model — it's useful for descriptive reporting (who's churning, from where, at what spend level) but not for confidently predicting churn from these features. Flagging this now so it doesn't get overstated in a portfolio writeup or interview.

Dashboard / Model / Output

- **Power BI dashboard:** `CHURN.pbix` — interactive report covering churn rate, satisfaction/NPS breakdowns, segment and regional revenue, and demographic splits.
- **SQL KPI layer:** `SQLQuery.sql` — the queries feeding the dashboard and any ad hoc business reporting.
- **No dashboard screenshots were included in what you gave me.** Export a few key pages from `CHURN.pbix` as PNGs, drop them in an `images/` folder in the repo, and reference them here (same pattern as the Sales Performance project README) — a GitHub visitor won't open a `.pbix` file just to see what the dashboard looks like.

Results and Conclusion

The dataset went from 50,300 raw records with multiple missing-value and formatting issues to 50,000 clean records, fully typed and loaded into SQL Server for KPI reporting. The analysis shows a churn rate of just over 30%, a customer base that's roughly gender-balanced, and revenue concentrated in the SMB segment and Subscription product category. Satisfaction and NPS scores move in the expected direction relative to churn (lower scores, more churn) but the relationship is weak — this dataset supports descriptive, segment-level churn reporting well, but would need additional behavioral or engagement features before it could support a real predictive churn model.

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